Import Data Using Transform Maps (Spreadsheet)

Phase 7 - Project Documentation Phase

1. Documentation Overview

The **Project Documentation Phase** focuses on recording all important information related to the **"Import Data Using Transform Maps (Spreadsheet)"** project.

The purpose of documentation is to provide a clear reference for the project's requirements, design, development, testing, implementation, and maintenance.

The documentation should explain how spreadsheet data is imported into ServiceNow, how the data is transformed, how records are created or updated, and how errors are handled.

2. Objectives of Documentation

The main objectives are:

- Document the complete project process.
- Record all functional and technical requirements.
- Explain the Import Set and Transform Map configuration.
- Document source-to-target field mappings.
- Document transformation rules.
- Record Coalesce configuration.
- Document validation and error-handling procedures.
- Record test cases and results.
- Provide implementation and usage instructions.
- Make future maintenance easier.
- Provide a reference for developers, administrators, testers, and users.

3. Project Documentation Structure

The complete project documentation can be organized into the following sections:

```
Project Documentation
        |
        ├── Project Overview
        ├── Business Requirements
        ├── Functional Requirements
        ├── System Design
        ├── Data Flow
        ├── Configuration Details
        ├── Field Mapping
        ├── Transformation Rules
        ├── Coalesce Configuration
        ├── Validation & Error Handling
        ├── Testing Documentation
        ├── Deployment Procedure
        ├── User Guide
        ├── Troubleshooting
        └── Maintenance & Support
```

4. Project Overview Documentation

The documentation should begin with a brief description of the project.

Project Title

Import Data Using Transform Maps (Spreadsheet)

Project Purpose

The purpose of this project is to provide a structured mechanism for importing spreadsheet data into ServiceNow using **Import Sets and Transform Maps**.

The solution allows users to upload spreadsheet data, temporarily store it in an Import Set Table, transform and validate the data, and finally insert or update records in the target ServiceNow table.

5. Business Requirement Documentation

The business requirement section should explain why the project is needed.

Business Problem

Manual entry of large amounts of spreadsheet data into ServiceNow can:

- Consume significant time.
- Increase data-entry errors.
- Create duplicate records.
- Produce inconsistent data.
- Require additional manual effort.

Proposed Solution

Use ServiceNow Import Sets and Transform Maps to automate the movement of spreadsheet data into the appropriate target table.

Expected Benefits

- Reduced manual effort.
- Faster bulk data import.
- Improved data accuracy.
- Better consistency.
- Duplicate handling.
- Easier data maintenance.

6. Functional Requirement Documentation

The documentation should record the major functional requirements.

Requirement	Description
Spreadsheet Upload	User should be able to upload the source spreadsheet
Import Set	Source data should be loaded into an Import Set

Requirement	Description
Import Set Table	Imported data should be temporarily stored
Transform Map	Source data should be mapped to the target table
Field Mapping	Source fields should map to correct target fields
Transformation	Data should be transformed when required
Coalesce	Existing records should be identified
Validation	Invalid or incomplete data should be identified
Error Handling	Failed records should be traceable
Target Update	New records should be inserted and existing records updated

7. System Design Documentation

The documentation should include the architecture of the solution.

System Flow

```
Spreadsheet
     ↓
Data Source
     ↓
Import Set
     ↓
Import Set Table
     ↓
Transform Map
     ↓
Field Mapping
     ↓
Transformation & Validation
     ↓
Coalesce Check
     ↓
Target Table
```

8. Data Source Documentation

The Data Source configuration should be documented.

The documentation should include:

- Data Source name.
- Source file type.
- File format.
- Import method.
- Import Set Table.
- Source spreadsheet structure.
- Required columns.

Example

Configuration	Description
Source	Spreadsheet
File Type	Excel/CSV
Import Method	File Upload
Import Set Table	Employee Import Table
Target	Employee Table

9. Import Set Documentation

The Import Set configuration should be documented.

The documentation should explain:

- Purpose of the Import Set.
- Source data structure.
- Import Set Table.

- Import execution procedure.
- Import status.
- Number of records imported.

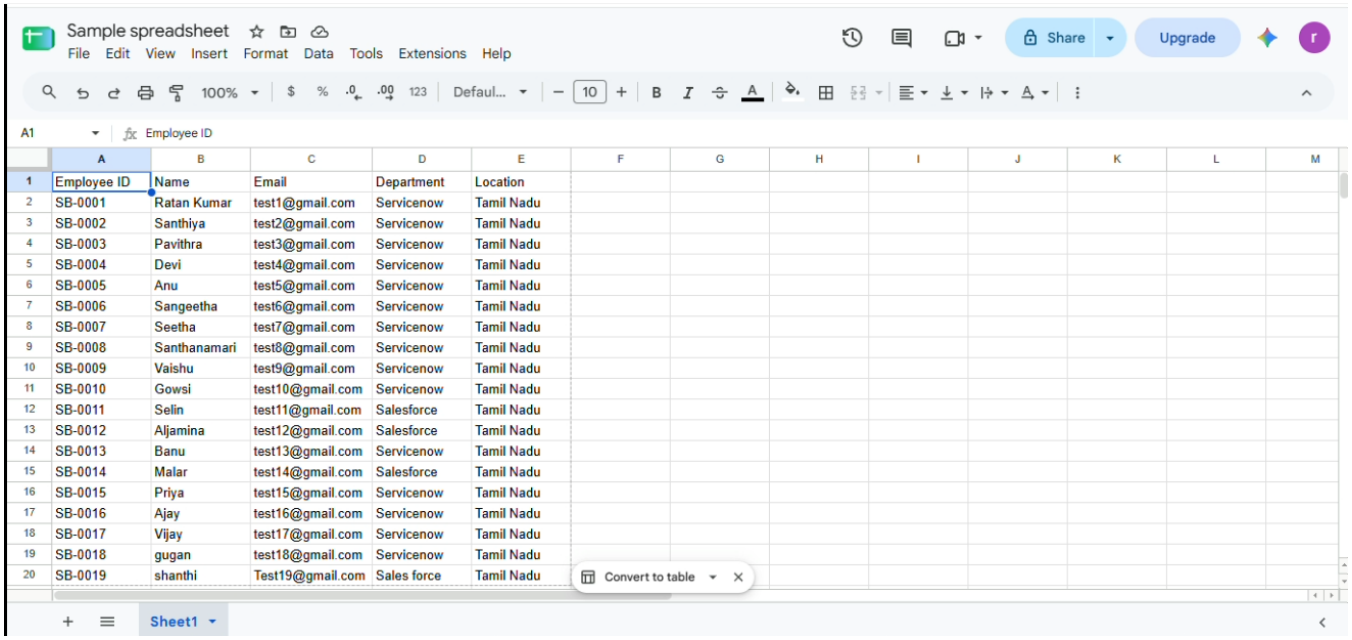
The Import Set acts as an intermediate layer between the spreadsheet and the target table.

10. Import Set Table Documentation

The Import Set Table structure should be documented.

Example:

Field	Data Type	Source
Employee ID	String	Spreadsheet
Employee Name	String	Spreadsheet
Email	String	Spreadsheet
Department	String	Spreadsheet
Joining Date	Date	Spreadsheet



	A	B	C	D	E
1	Employee ID	Name	Email	Department	Location
2	SB-0001	Ratan Kumar	test1@gmail.com	Servicenow	Tamil Nadu
3	SB-0002	Santhiya	test2@gmail.com	Servicenow	Tamil Nadu
4	SB-0003	Pavithra	test3@gmail.com	Servicenow	Tamil Nadu
5	SB-0004	Devi	test4@gmail.com	Servicenow	Tamil Nadu
6	SB-0005	Anu	test5@gmail.com	Servicenow	Tamil Nadu
7	SB-0006	Sangeetha	test6@gmail.com	Servicenow	Tamil Nadu
8	SB-0007	Seetha	test7@gmail.com	Servicenow	Tamil Nadu
9	SB-0008	Santhanamari	test8@gmail.com	Servicenow	Tamil Nadu
10	SB-0009	Vaishu	test9@gmail.com	Servicenow	Tamil Nadu
11	SB-0010	Gowsi	test10@gmail.com	Servicenow	Tamil Nadu
12	SB-0011	Selin	test11@gmail.com	Salesforce	Tamil Nadu
13	SB-0012	Aljamina	test12@gmail.com	Salesforce	Tamil Nadu
14	SB-0013	Banu	test13@gmail.com	Servicenow	Tamil Nadu
15	SB-0014	Malar	test14@gmail.com	Salesforce	Tamil Nadu
16	SB-0015	Priya	test15@gmail.com	Servicenow	Tamil Nadu
17	SB-0016	Ajay	test16@gmail.com	Servicenow	Tamil Nadu
18	SB-0017	Vijay	test17@gmail.com	Servicenow	Tamil Nadu
19	SB-0018	gugan	test18@gmail.com	Servicenow	Tamil Nadu
20	SB-0019	shanthi	Test19@gmail.com	Sales force	Tamil Nadu

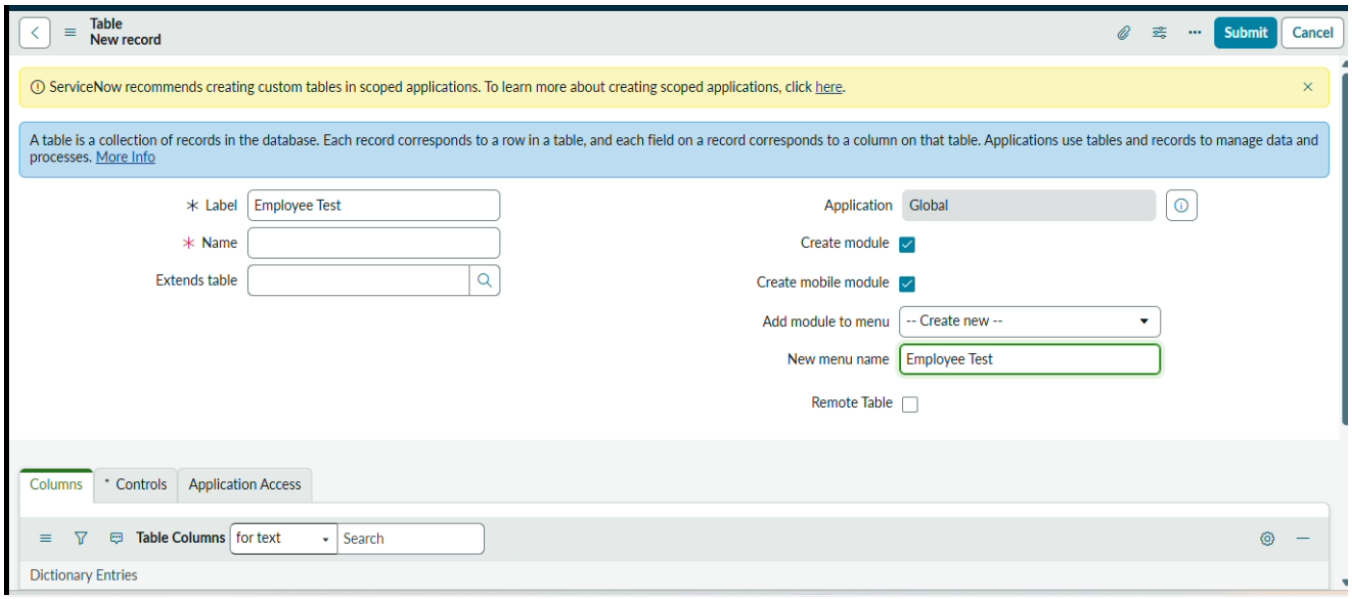
The documentation should clearly identify the purpose of each field.

11. Transform Map Documentation

The Transform Map is the primary transformation component and should be documented in detail.

The documentation should include:

- Transform Map name.
- Source table.
- Target table.
- Field mappings.



- Coalesce fields.
- Transformation logic.
- Scripts, if used.
- Execution order where applicable.
- Error-handling behavior.

Example

Transform Map: Employee Spreadsheet Transform

Source: Employee Import Set Table

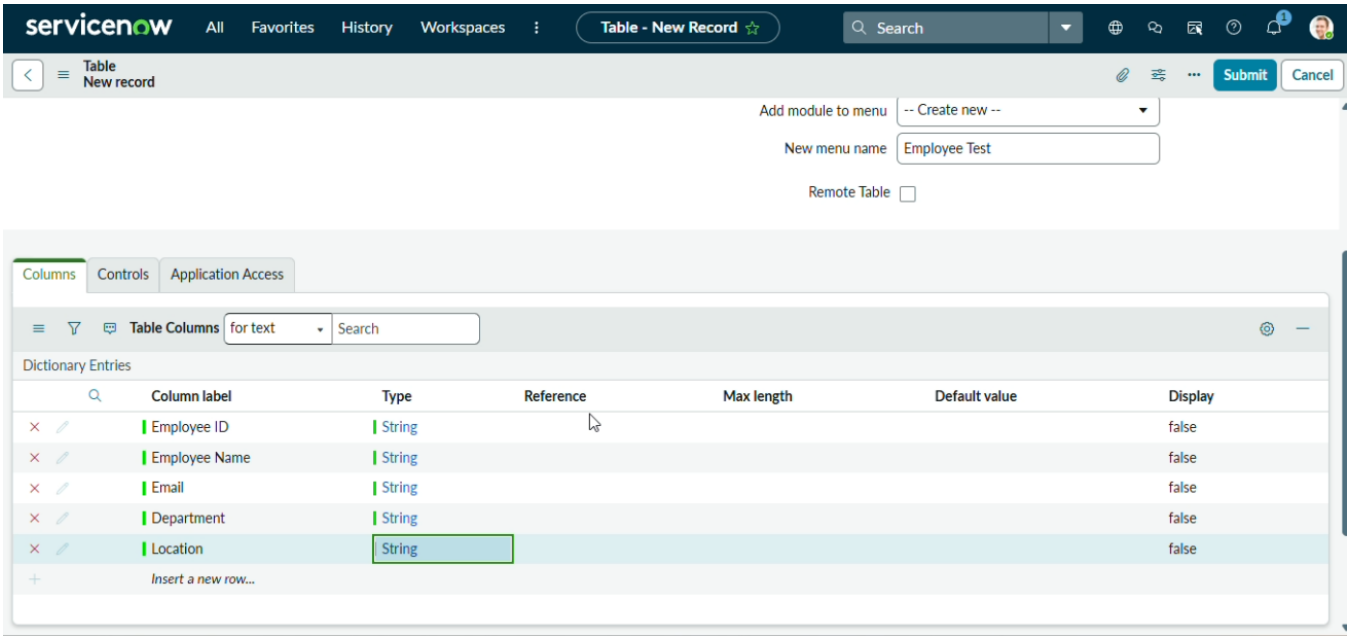
Target: Employee Table

12. Field Mapping Documentation

The source-to-target mapping should be clearly documented.

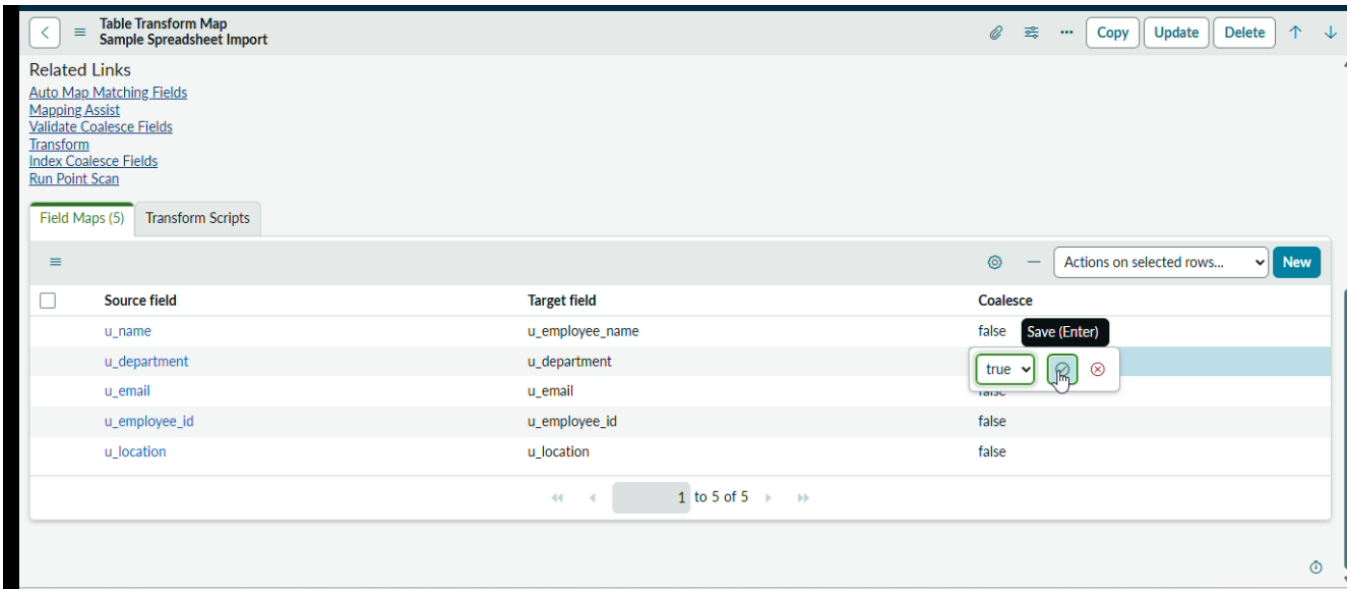
Source Field	Target Field	Transformation
Employee ID	Employee ID	Direct Mapping
Employee Name	Name	Direct Mapping
Email	Email	Direct Mapping
Department	Department	Reference/Value Mapping
Joining Date	Date of Joining	Date Conversion

This mapping table helps developers and administrators understand how data moves between tables.



13. Coalesce Documentation

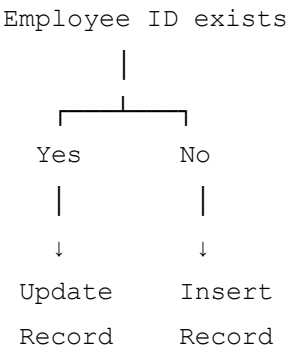
The documentation should explain how duplicate handling is configured.



For example:

Coalesce Field: Employee ID

Logic



This ensures that existing records are updated instead of creating unnecessary duplicates.

14. Transformation Logic Documentation

All transformation rules should be documented.

Examples:

- Date format conversion.
- Text standardization.
- Department value mapping.
- Empty value handling.
- Reference field lookup.
- Data cleanup.

If a Transform Script is used, the documentation should describe:

- Purpose of the script.
- Input values.
- Processing logic.
- Expected output.
- Conditions under which the script executes.

15. Validation Documentation

The documentation should define the validation rules.

Example:

Validation	Expected Behavior
Employee ID missing	Record should be flagged
Employee Name missing	Record should be flagged
Invalid email	Validation error
Invalid date	Transformation/validation error
Invalid department	Record should be handled according to business rule
Duplicate Employee ID	Coalesce logic applied

This provides a clear reference for data-quality requirements.

16. Error Handling Documenttion

The project documentation should explain how errors are identified and resolved.

Common Errors

- Missing mandatory fields.
- Invalid data format.
- Incorrect field mapping.
- Invalid reference values.
- Transformation failure.
- Duplicate/conflicting records.

Error Resolution Process

```
Identify Error
      ↓
```

```
Review Import Result
      ↓
Find Incorrect Data
      ↓
Correct Spreadsheet
      ↓
Re-import Data
      ↓
Execute Transform Map
      ↓
Verify Result
```

17. Testing Documentation

All testing activities should be recorded.

The documentation should include:

- Test case ID.
- Test scenario.
- Test input.
- Expected result.
- Actual result.
- Test status.
- Defect details, if applicable.

Example:

Test ID	Scenario	Expected Result	Status
TC01	Valid spreadsheet	Records imported	Pass
TC02	New Employee ID	New record created	Pass
TC03	Existing Employee ID	Record updated	Pass
TC04	Missing mandatory field	Error identified	Pass

Test ID	Scenario	Expected Result	Status
TC05	Invalid date	Error identified	Pass
TC06	Duplicate ID	Duplicate handled	Pass

18. Deployment Documentation

The deployment procedure should be documented so that the solution can be moved from the development environment to the required environment.

The deployment documentation may include:

1. Verify configuration in the development instance.
2. Complete testing.
3. Review Transform Map configuration.
4. Verify field mappings.
5. Verify security permissions.
6. Move the required configuration.
7. Perform post-deployment testing.
8. Verify target records.

19. User Guide

A simple user guide should be provided for users who need to perform spreadsheet imports.

User Procedure

1. Prepare the spreadsheet using the approved template.
2. Verify all mandatory fields.
3. Check the data for errors or duplicates.
4. Upload the spreadsheet.
5. Run the Import Set.
6. Execute the Transform Map.

7. Review the import results.
8. Check the target records.
9. Correct and re-import failed records if necessary.

This ensures that users can perform the import process consistently.

20. Troubleshooting Documentation

A troubleshooting section should be included.

Problem	Possible Cause	Solution
Spreadsheet fails to import	Incorrect file format	Use the approved format
Records not created	Transform Map issue	Verify Transform Map
Wrong field populated	Incorrect mapping	Review field mapping
Duplicate records	Coalesce not configured correctly	Verify unique field
Transformation failure	Invalid source data	Correct spreadsheet data
Reference not found	Invalid reference value	Verify reference data
Mandatory field error	Missing required value	Populate required field

21. Maintenance Documentation

The documentation should explain how the solution can be maintained in the future.

Maintenance activities may include:

- Updating field mappings.
- Adding new spreadsheet columns.
- Modifying transformation rules.
- Updating validation logic.
- Changing the target table.
- Updating Coalesce fields.

- Reviewing import errors.
- Updating the spreadsheet template.

Any configuration change should be tested before being applied to the production environment.

22. Version Control

The project documentation should maintain version information.

Example:

Version	Date	Change	Author
1.0	Initial	Initial project documentation	Project Team
1.1	Updated	Field mapping updated	Project Team
1.2	Updated	Validation rules updated	Project Team

This helps track changes made throughout the project lifecycle.

23. Project Deliverables Documentation

The final documentation package should contain:

- Project overview.
- Requirement document.
- Project design document.
- Development document.
- Configuration details.
- Spreadsheet template.
- Field mapping document.
- Transform Map details.
- Test cases.
- Test results.
- Error-handling procedure.

- Deployment procedure.
- User guide.
- Troubleshooting guide.
- Maintenance guide.

24. Documentation Completion Criteria

The documentation phase will be considered complete when:

- All project requirements are documented.
- System architecture is documented.
- Data flow is documented.
- Data Source configuration is documented.
- Import Set configuration is documented.
- Transform Map configuration is documented.
- Field mappings are documented.
- Coalesce logic is documented.
- Transformation rules are documented.
- Validation rules are documented.
- Test cases and results are recorded.
- Known errors and solutions are documented.
- Deployment steps are documented.
- User instructions are completed.
- Maintenance procedures are documented.

25. Final Project Documentation Flow

```
Project Overview
       ↓
Requirements Documentation
       ↓
System Design Documentation
       ↓
Configuration Documentation
```

```
        ↓
Field Mapping Documentation
        ↓
Transformation Documentation
        ↓
Validation & Error Handling
        ↓
Testing Documentation
        ↓
Deployment Documentation
        ↓
User Guide
        ↓
Troubleshooting Guide
        ↓
Maintenance Documentation
        ↓
Final Project Documentation
```

26. Final Project Documentation Statement

The Project Documentation Phase provides a complete reference for the “Import Data Using Transform Maps (Spreadsheet)” solution. It documents the project requirements, system architecture, Data Source, Import Set, Import Set Table, Transform Map, field mappings, transformation rules, Coalesce configuration, validation, error handling, testing, deployment, and maintenance procedures. The documentation ensures that the developed solution can be understood, operated, tested, maintained, and enhanced by future developers, administrators, testers, and business users.